

Department of CSE (Data Science)

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Subject: Business Intelligence (23UDSPEL4704) — TAE 1

REPORT TITLE

Cleaning and Analyzing the Raw India Post Office Directory to Improve Postal Delivery Coverage and Service-Gap Intelligence

1. Student Credentials

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1. Problem Statement & Business Value

India Post runs the largest postal network in the world — around 1.55 lakh post offices spread across every state and union territory, each one classified as a Head Office, Sub Office, or Branch Office, and marked as either a Delivery or Non-Delivery point. This directory is available as open data on data.gov.in and can also be pulled through a live REST API. Separately, India Post's public tracking tool lets anyone look up the journey of a single parcel. The problem is that these pieces have never really been put together. You can see where post offices are, and you can check on one parcel at a time, but there's no view that connects the two — nothing that tells you whether a region with fewer delivery points also happens to be a region where parcels take longer to arrive.

Business Domain: Postal Logistics / Last-Mile Delivery Business Intelligence.

Key Operational Bottlenecks

- There's no combined view linking how many post offices/delivery points a region has to how long parcels actually take to reach that region.
- The post office directory has no unique ID for each record and no clean hierarchy, so getting a reliable circle-level or state-level count means building that structure from scratch.
- Delivery status and location fields aren't filled in consistently — some coordinates are missing (and not even marked as missing properly), and status labels vary — so basic coverage numbers can't be trusted straight out of the source.
- There's no bulk parcel-transit dataset — the tracking tool only shows one consignment at a time — so any delay analysis has to start from a manually built sample.

Target Decision-Makers

- Postal circle/regional administrators trying to decide where to add delivery infrastructure.
- Department of Posts policy planners looking at whether last-mile service is equally good in rural and urban areas.
- Logistics/e-commerce companies that depend on India Post for last-mile delivery in areas they don't cover themselves.
- Ordinary customers and small businesses who just want a realistic idea of how long delivery will take in their area.

Target KPIs to be enabled: post office density and delivery-point ratio by district/region; average parcel transit time (booking to delivery) by circle; delay rate against expected service levels; an underserved-region index (low delivery-point density combined with high transit delay); and a data-completeness KPI (share of records with valid, non-missing geo-coordinates).

2. Literature & Industry Survey

Before finalizing this problem statement, we looked at a few recent papers on postal and last-mile delivery performance, mainly to check whether the business angle we're proposing actually holds up and whether others have run into the same data problems:

- A 2024 systematic review in *Frontiers in Future Transportation* looked at how the spatial layout of a delivery network affects service performance, and found that areas with more delivery points nearby consistently perform better than sparser ones. This lines up with our core assumption — that post office density should have some relationship with how fast parcels move.
- A study published in the *International Journal of Physical Distribution & Logistics Management* looked at last-mile delivery inequity across an entire country using postal-style survey data, and found that service quality varies a lot by region rather than being uniform.

This is useful for us because it backs up why a single national average KPI isn't enough — we need region-level numbers.

- A paper on predicting parcel delivery times using deep learning, built on real data from a national postal operator, showed that once raw operational logs are properly structured, delivery-time prediction becomes a realistic, well-supported problem. That's basically the same path we're planning to take with our sampled tracking data, just starting from the BI/warehousing stage rather than jumping straight to modelling.

3 Dataset Identity

Dataset Name: “All India Pincode Directory till Last Month.”

Publishing Authority: Ministry of Communications, Department of Posts, Government of India, via data.gov.in.

Dataset Page: https://www.data.gov.in/datasets_webservices/datasets/6818292

License: Released under the National Data Sharing and Accessibility Policy (NDSAP) of the Government of India.

Supplementary raw sources: India Post's public consignment-tracking tool (per-article status logs, manually sampled) and the Department of Posts Annual Report (mail volume/performance tables published as PDF).

4. Sourcing Proof

- We pulled this straight from the live government API (resource ID 5c2f62fe-5afa-4119-a499-fec9d604d5bd) — not from any third-party site or a repackaged repository.
- The field names come exactly as the source system stores them — all lower-case, no spacing, nothing renamed for readability (circlename, officename, statename). Nobody has cleaned this up for presentation.
- Missing values show up as the literal text “NA” instead of an actual blank/null — which by itself is proof this hasn't been through any cleaning pass yet.
- The parcel-tracking history isn't even downloadable as a file — it only exists one consignment at a time through the live tracking tool, so we had to build our own small sample manually. It's hard to get more raw than that.

5. Dataset Profile

Metric	Observed Value
Total post offices (India-wide directory)	~1,55,000 (India Post: world's largest postal network)
States covered	36
Postal circles	23
Fields per record	circlename, regionname, divisionname, officename, pincode, officetype, delivery, district, statename, latitude, longitude
Delivery status field	'Delivery' / 'Non Delivery' (non-standardized labels)
Geo-coordinate completeness	Partial — latitude/longitude present as literal string 'NA' for a meaningful share of branch offices

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- Missing values show up as the literal text “NA” instead of an actual blank/null — which by itself is proof this hasn't been through any cleaning pass yet.

- There's no target/label column, no train-test split, and no “cleaned dataset” documentation anywhere — none of the things you'd expect from a tutorial-ready CSV.
- The parcel-tracking history isn't even downloadable as a file — it only exists one consignment at a time through the live tracking tool, so we had to build our own small sample manually. It's hard to get more raw than that.

6. Identified Raw Data Issues

Issue	Description (observed in the sourced API response)	Business / Analytical Impact
Missing geo-coordinates disguised as text	latitude and longitude fields contain the literal string "NA" instead of null/blank for many Branch Offices, e.g. Kammergaon B.O (pincode 504296).	Breaks numeric processing and map-based visualization unless explicitly type-checked and handled before geospatial analysis.
Inconsistent free-text spacing	officename values contain irregular internal spacing, e.g. "Kammergaon B.O" (double space before suffix).	Naive string joins/lookups on office name will silently fail to match otherwise-identical entries.
Non-standardized delivery status labels	delivery field observed as "Delivery" and "Non Delivery" (no hyphen, inconsistent casing/spacing risk across the full 1.5 lakh-row dataset).	Category counts and coverage KPIs will undercount/overcount unless label variants are normalized first.
No native unique row identifier	No single field uniquely identifies a record; pincode alone is not unique (multiple offices share the same pincode, e.g. 504296).	Requires a constructed composite key (Circle + Division + Office + Pincode) before this data can be safely warehoused — a genuine 'unindexed keys' problem.
Fragmented hierarchy with no ID fields	Circle, Region, Division, District, and State are stored only as repeated text strings, not as normalized foreign keys.	Rollups (state-level, division-level coverage) require building a proper dimensional hierarchy from scratch.
Bulk transit/tracking data not publicly available	The public tracking tool exposes only per-consignment lookup (one tracking number at a time); no bulk historical tracking feed exists.	Delivery-time KPIs must initially rely on a manually sampled batch of tracking numbers rather than a large automated feed — a genuine data-collection constraint to be documented and addressed via scheduled manual/API sampling in Phase II.

7. Proposed Methodology & Architecture

The plan is to pull the raw post office directory through the API, combine it with our manually sampled tracking log, clean and profile both, load them into a dimensional warehouse, and then put a dashboard on top of it. We're keeping ingestion, cleaning, warehousing, and reporting as separate stages on purpose.

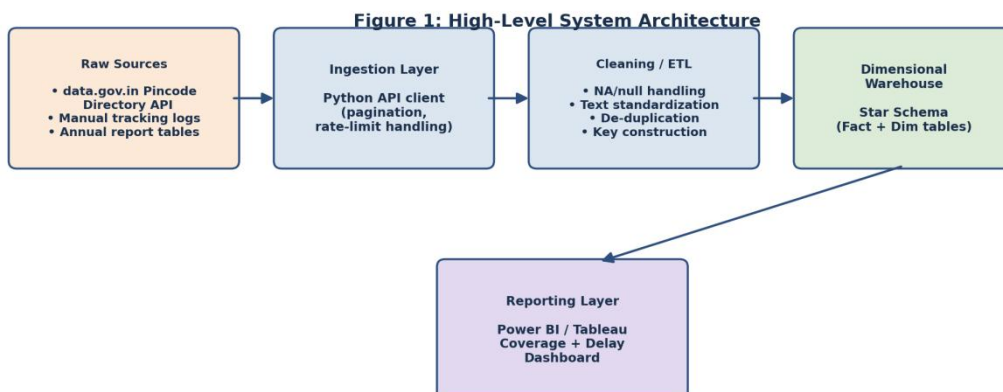


Figure 1: High-level system architecture, from raw API ingestion through to the reporting layer.

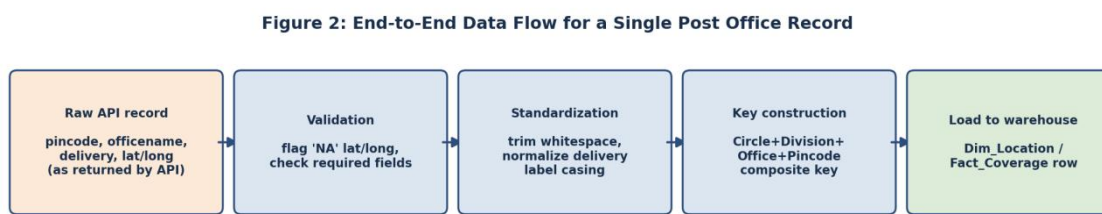


Figure 2: End-to-end data flow for a single post office record.

7.1 Data Cleaning / ETL Plan

- Fix the missing values: turn the “NA” strings in latitude/longitude into actual nulls and flag whichever records don't have usable coordinates.
- Clean up the text: strip out the extra whitespace in office names and other text fields, and get the delivery status down to one consistent set of labels instead of whatever variants show up in the raw data.
- Build a proper key: since there's no primary key, we'll construct one ourselves from Circle + Division + Office Name + Pincode.
- Normalize the hierarchy: turn the repeated Circle/Region/Division/District/State text into real dimension tables with their own keys, instead of just strings repeated on every row.
- Bring in the tracking sample: structure our manually collected tracking log (booking time, status updates, delivery time) into a consistent format, sorting out the timestamp inconsistencies along the way.
- Check referential integrity: make sure every Branch/Sub Office actually maps to a real Division and Circle, and flag anything that doesn't.

7.2 Target Schema Design — Star Schema

Fact Table — Fact_Coverage: Office_Key, Location_Key, Date_Key, Delivery_Status, Has_Valid_Geo (derived).

Fact Table — Fact_Delivery_Sample: Consignment_Key, Origin_Location_Key, Destination_Location_Key, Date_Key, Transit_Hours (derived: Delivered_Time – Booked_Time).

Dimension Tables:

- Dim_Location — Circle, Region, Division, District, State.
- Dim_Office — Office Name, Office Type (H.O/S.O/B.O), Pincode, Latitude, Longitude, Delivery Status.
- Dim_Date — Date, Day, Week, Month, Quarter, Year.

A Star Schema is proposed over Snowflake for Phase II because the dimension tables here are shallow (no deep sub-hierarchies beyond the Circle→Region→Division→District chain, which collapses cleanly into a single Dim_Location table), keeping query performance high for dashboard-level aggregation — the primary Phase II deliverable.

7.3 Primary Tools

- Python (pandas, requests) — API ingestion, pagination handling, and cleaning pipeline.
- SQL (PostgreSQL/MySQL) — dimensional data warehouse implementation.
- Power BI / Tableau — coverage and delay dashboard for Phase II.
- data.gov.in REST API — scheduled raw directory ingestion; India Post tracking tool — manual delivery-sample collection.

8. Conclusion

The India Post directory and our manually sampled tracking data give us a large, credible, and genuinely messy government-linked source — exactly the kind of raw data this proposal set out to find. It has all the real problems: missing values hidden as text, delivery labels that aren't standardized, no primary key to rely on, and no bulk history of parcel movement to pull from. Once this is cleaned and modelled into a proper warehouse, it gives postal administrators and policy planners one dependable place to look at coverage and delivery performance together — a business case with real stakeholders behind it, and a clear path from raw ingestion in Phase I to a working dashboard in Phase II.